



COMP47350: Data Analytics (Conv)

Dr. Georgiana Ifrim

georgiana.ifrim@ucd.ie

Insight Centre for Data Analytics

ML-Labs Centre for Research Training in Machine Learning

School of Computer Science

University College Dublin

2025/26

Module Topics

- Python Environment (Anaconda, Jupyter Notebook)
- Getting Data (Web scraping, APIs, DBs)
- **Understanding Data (stats, visualisation)**
- **Preparing Data (cleaning, transformation)**
- **Modeling & Evaluation (machine learning)**

Data Understanding & Preparation



Monica Rogati

@mrogati

 Follow



Hey, data scientists: data prep is not beneath you. Good data prep is detective work & takes intuition, experience, ingenuity & pragmatism.

Case Study



Motor Insurance Claim Fraud Detection

Business problem:

Motor insurance company is losing money due to **fraudulent claims**.

Analytics solution:

Claim Prediction: Is a new claim likely to be fraudulent or not?

Data Understanding

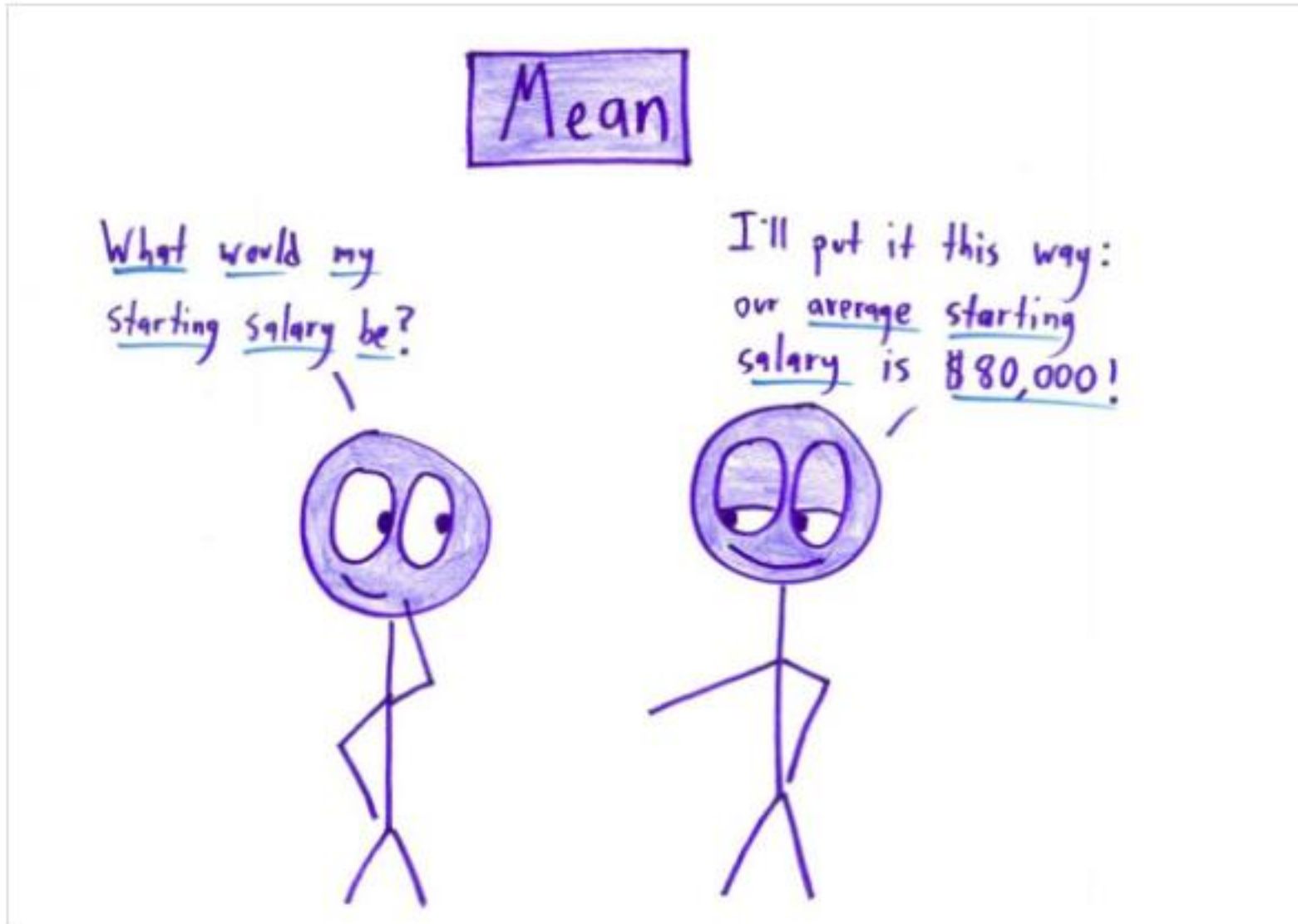
- **Data Quality Report**
 - Getting To Know The Data
 - Identifying Data Quality Issues
- **This lecture: diving into descriptive statistics**

(a) Continuous Features

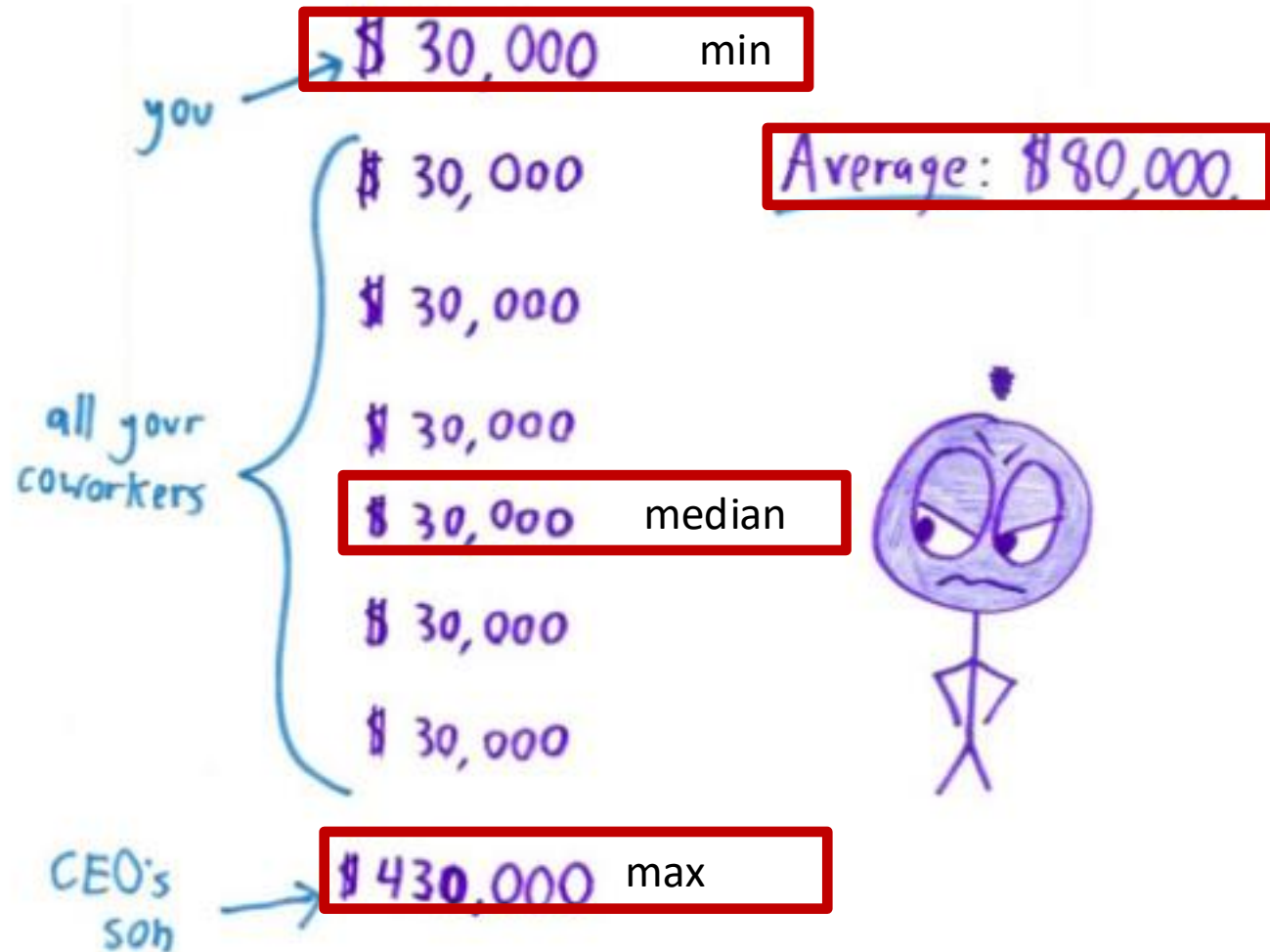
(b) Categorical Features

[illegible]

Why Not **Blindly** Trust Statistics



Why Not Blindly Trust Statistics



Continuous Features: Descriptive Stats

- The **mean** and **standard deviation** of each feature give a sense of the **central tendency** and **variation** of the values within the dataset
- The **minimum** and **maximum** values give the **range** that is possible for each feature
[min,....., max]

Continuous Features: Mean

Arithmetic mean (or **sample mean**) of a set of n values for a feature $\mathbf{a}$, denoted as $a_1, a_2, \dots, a_n$, is denoted by the symbol $\bar{a}$, and is calculated as (sum of values, divided by number of samples):

$$\bar{a} = \frac{1}{n} \sum_{i=1}^n a_i$$

Continuous Features: Mean

Example

ID	1	2	3	4	5	6	7	8
Height	150	163	145	140	157	151	140	149

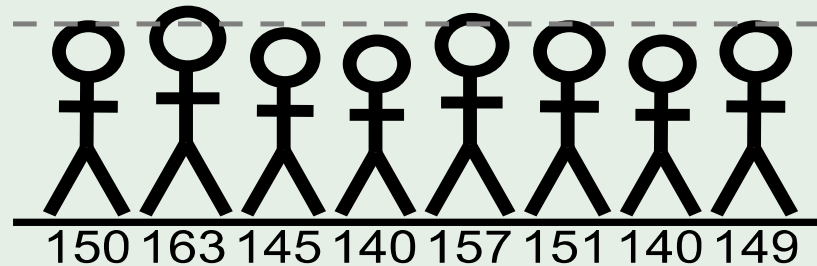


Figure: The members of a school basketball squad. The dashed grey line shows the arithmetic mean of the players' heights.

$$\begin{aligned}\overline{\text{HEIGHT}} &= \frac{1}{8} \times (150 + 163 + 145 + 140 + 157 + 151 + 140 + 149) \\ &= 149.375\end{aligned}$$

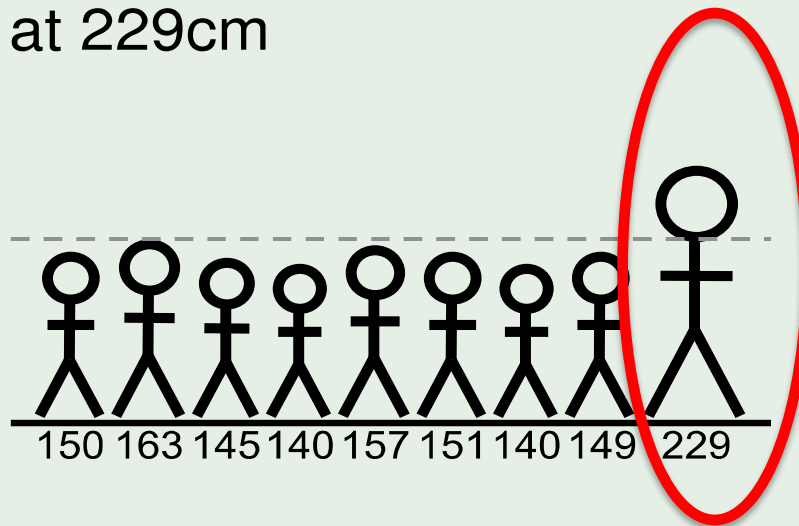
Continuous Features: Mean

- The **arithmetic mean** is one measure of the **central tendency of a sample**
- Any measure of **central tendency** is, however, just an **approximation**, which can be affected by extreme values (also known as **outliers**)

Continuous Features: Outliers Effect on the Mean

Example

- Suppose our basketball squad manage to sign a *ringer* measuring in at 229cm



- The arithmetic mean for the full group is 158.235cm and no longer represents the central tendency of the group.
- An unusually large or small value like this is referred to as an **outlier** - the arithmetic mean is very sensitive to outliers.

Continuous Features: Range

- The simplest measure of variation is the **range**: $\text{range}(a) = \max(a) - \min(a)$
- For our previous example:
 $\text{range}(\text{Height}) = 229 - 140 = 89\text{cm}$
- Another measure of variation is the deviation or the **variance around the mean**

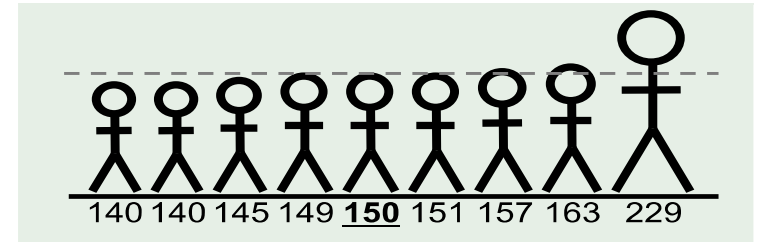
Continuous Features: Variance

- The **variance** of a sample of values is the **average difference between each value in a sample and the mean of that sample**
- The **variance** of n values of a feature $\mathbf{a}$, *i.e.*, $a_1, a_2, \dots, a_n$, is denoted **$\text{var}(\mathbf{a})$** and is calculated as:

$$\text{var}(\mathbf{a}) = \frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n - 1}$$

Continuous Features: Variance

- Variance example: $n = 9$



Mean=158.23

$$\begin{aligned} \text{var}(\text{height}) &= \frac{1}{9-1} ((140 - 158.23)^2 + (140 - 158.23)^2 \\ &\quad + (145 - 158.23)^2 + (149 - 158.23)^2 \\ &\quad + (150 - 158.23)^2 + (151 - 158.23)^2 \\ &\quad + (157 - 158.23)^2 + (163 - 158.23)^2 \\ &\quad + (229 - 158.23)^2) = 759.69 \end{aligned}$$

$$\text{var}(a) = \frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n - 1}$$

Variance is not measured in the original units, i.e., 759.69 are not measured in cm, because we squared the deviation of each value from the mean

Continuous Features: Standard Deviation

- The standard deviation of a sample, denoted by *sd*, is calculated by taking the square root of the **variance** of the sample (this way we go back to the original units, e.g., cm):

$$\begin{aligned} sd(a) &= \sqrt{var(a)} \\ &= \sqrt{\frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n - 1}} \end{aligned}$$

Continuous Features: Standard Deviation

- Standard deviation example (back to original units, e.g., standard deviation is 27.56 cm):

$$sd(height) = \sqrt{var(height)} = \sqrt{759.69} = 27.56$$

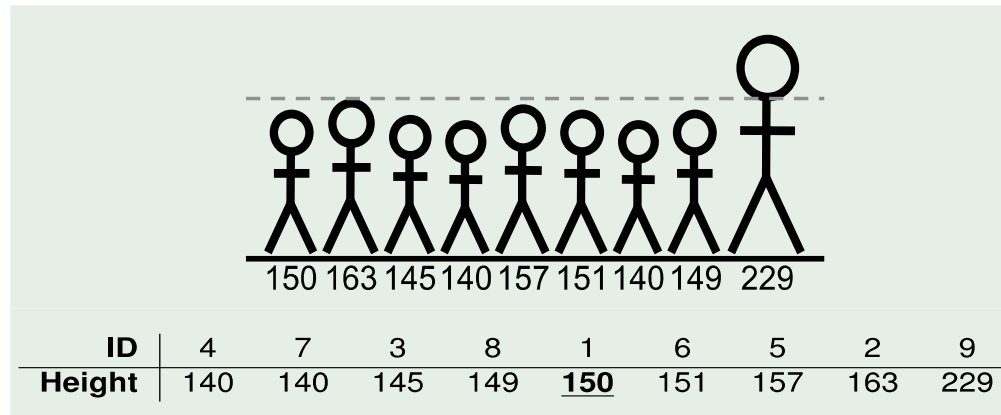
$$sd(a) = \sqrt{var(a)}$$

$$= \sqrt{\frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n - 1}}$$

Standard deviation is measured in the original units, i.e., **27.56 cm**

Continuous Features: Descriptive Statistics

- For our example sample: 9 basketball players and their heights



- Mean(height) = 158.23; Sd(height) = 27.56**
- Often, we use **mean $\pm$ sd** to describe a sample: 158.23 ± 27.56
- Human height is Normal distributed (aka Gaussian or bell shape distributed): it is rare to have very small heights or very big heights; most people have a height within 160cm to 180cm
- For the Gaussian distribution, about 99% of values fall within **mean $\pm$ 3 sd**

Continuous Features: Median

- **Median**: the middle value in a set of **ordered** values (from lowest to highest)
- If there is an even number of values in the sample, then the median can be obtained by calculating the arithmetic mean of the middle two values
- Half of the values are smaller (or equal) and half are bigger (or equal) than the median
- **50% of values in the sample are $\leq$ median of the sample**
- Median is also called the 50th percentile
- Note: there are several ways to compute the median as long as the property of the median holds (e.g., is the median a value in the set or not, how is the median approximated)

Continuous Features: Median

Example

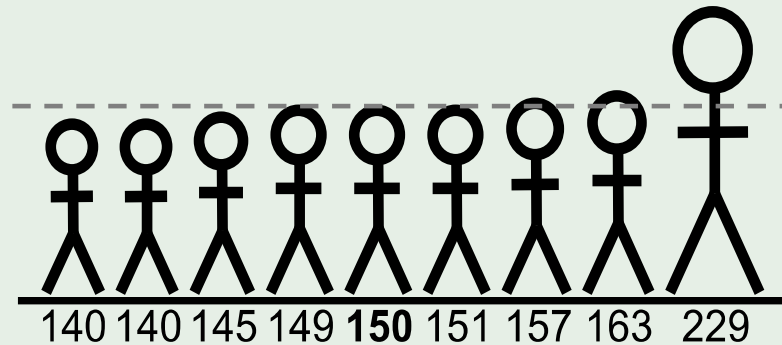


Figure: The members of the school basketball squad ordered by height, the dashed grey line shows the **median**.

ID	4	7	3	8	1	6	5	2	9
Height	140	140	145	149	<u>150</u>	151	157	163	229

Continuous Features: Percentiles

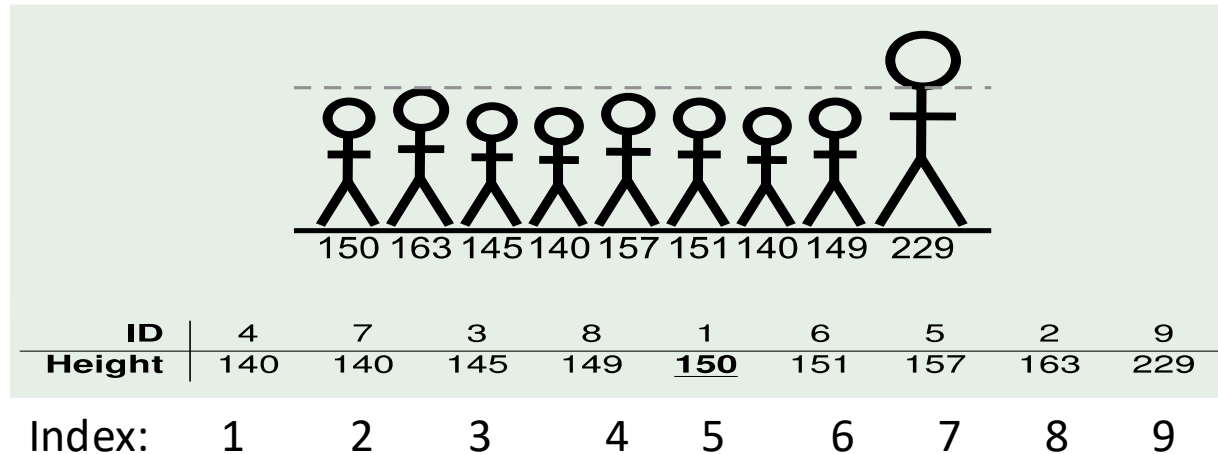
- **Percentiles** measure the variation of the values for a feature (values sorted from min to max value). They give more information than the mean and deviation and are less susceptible to outliers.
- **i^{th} percentile:** A proportion of $i/100$ of the values in a sample take values equal to or lower than the i^{th} percentile of that sample
- Median = 50th percentile (also called 2nd quartile); **50% of values are $\leq$ than the median value**
- **25th percentile:** 25% of the values fall $\leq$ than this (also called the **1st quartile**)
- **75th percentile:** 75% of the values fall $\leq$ than this (also called the **3rd quartile**)

Continuous Features: Percentiles

- We can use percentiles to describe another **measure of variation** known as the **inter-quartile range**
- The inter-quartile range is calculated as the difference between the **25th percentile** and the **75th percentile**
- These percentiles are also known as the **lower quartile** (or 1st quartile) and **upper quartile** (or 3rd quartile) hence the name **inter-quartile range**.

Continuous Features: Percentiles

For our example:



What is the **25th** percentile for Height? (computed as in the FMLPDA textbook)

$(25/100) * 9 = 2.25$ (9 is the number of players, 2.25 is an index)

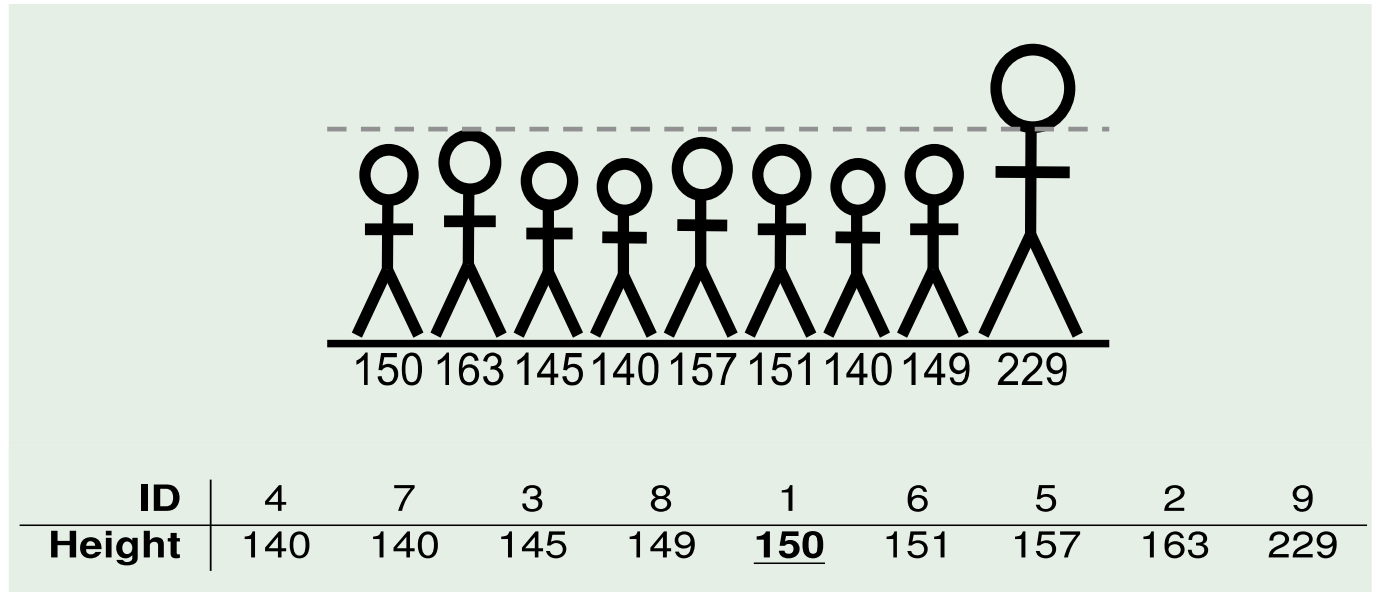
Interpolate between the values at index 2 (i.e., 140) and index 3 (i.e., 145) in the sorted order: $(1-0.25)*140 + (0.25)*145 = 105 + 36.25 = \mathbf{141.25}$

What is the 75th percentile? (155.5)

- An alternative way to compute percentiles is to round up to the next index, this way the percentile is an actual value of our feature: 25th perc = 145 (value at index 3), 75th perc = 157 (value at index 7). Numpy calculates this way.

Continuous Features: Percentiles

For our
example:



- Summary statistics for feature Height:

Feature	Min	1 st Quartile	Mean	Median	3 rd Quartile	Max	Std Dev
Height	140	141.25	158.23	150	155.5	229	27.56

Continuous Features: Percentiles

Computing percentiles in Python using numpy. There are different ways to compute percentiles (in this simple example, different ways give the same results, eg linear, lower, etc).

```
a = np.array([140, 140, 145, 149, 150, 151, 157, 163, 229])
print("a: " + str(a))

min_a = np.min(a)
print("min: " + str(min_a))

p_25 = np.percentile(a, 25)
print("1st q:" + str(p_25))

median_a = np.median(a)
print("median: " + str(median_a))

p_50 = np.percentile(a, 50)
print("2nd q:" + str(p_50))

p_75 = np.percentile(a, 75)
print("3rd q:" + str(p_75))

max_a = np.max(a)
print("max: " + str(max_a))

mean_a = np.mean(a)
print("\nmean: " + str(mean_a))
```

```
a: [140 140 145 149 150 151 157 163 229]
min: 140
1st q:145.0
median: 150.0
2nd q:150.0
3rd q:157.0
max: 229
```

```
mean: 158.22222222222223
```

```
p_25_linear = np.percentile(a, 25, interpolation='linear')
print("25perc linear:" + str(p_25_linear))
```

```
p_25 = np.percentile(a, 25, interpolation='lower')
print("25perc lower:" + str(p_25))
```

```
p_25 = np.percentile(a, 25, interpolation='higher')
print("25perc higher:" + str(p_25))
```

```
p_25 = np.percentile(a, 25, interpolation='nearest')
print("25perc nearest:" + str(p_25))
```

```
p_25 = np.percentile(a, 25, interpolation='midpoint')
print("25perc midpoint:" + str(p_25))
```

```
25perc linear:145.0
25perc lower:145
25perc higher:145
25perc nearest:145
25perc midpoint:145.0
```

Lab Today

- Practice data understanding and stats (i.e., Data Quality Report/Tables) using the dataset for the **Motor Insurance Case Study** (and Python libraries **pandas**, **matplotlib**)
- Lab3-Notebook-DataUnderstanding-MotorInsuranceData.ipynb

References

- **FMLPDA Book: Fundamentals of Machine Learning for Predictive Data Analytics**, John D. Kelleher, Brian Mac Namee, Aoife D'Arcy, MIT Press, 2015 (<http://machinelearningbook.com/>)
- **Why Not to Trust Statistics | Math with Bad Drawings** (Twitter: @benorlin)
<https://mathwithbaddrawings.com/2016/07/13/why-not-to-trust-statistics/#more-4815>
- **Data Exploration with Python**
<http://blog.districtdatalabs.com/data-exploration-with-python-part-2--2>
- **Engaging Data Stories** (Hans Rosling):
<https://www.indy100.com/article/hans-rosling-death-ted-talk-data-7568656>